

A Multi-Scale Deep Learning Approach for Hairline Crack Detection using a Hybrid CNN–Transformer Model

Sarita Sapkota¹, Ravi Kumar Pajiyar², and Prof. Dr. Sudan Jha²

¹Department of Civil Engineering, Kathmandu University

²Department of Computer Science & Engineering, Kathmandu University
sudan.jha@ku.edu.np, ss44041320@student.ku.edu.np,
rp33021320@student.ku.edu.np

Abstract. Detecting hairline cracks, often less than 0.3 mm wide, is a critical but difficult task in structural health monitoring. These cracks are faint, easily lost in surface noise, and hard to segment accurately. To address this, we developed **CrackFormer**, a new hybrid CNN-Transformer model. Our approach uses a multi-scale CNN to capture fine local details and a Swin Transformer to understand the overall continuity of a crack, even if it's broken or faint. We also use attention gates to help the model focus on crack features and ignore distracting background patterns. We tested CrackFormer on a challenging dataset combining SDNET2018, DeepCrack, and 1,200 of our own field images. It achieved an IoU of **85.4%** and an F1-score of **92.1%**, surpassing current models like U-Net and TransUNet by 3-5%. Finally, we show how our model can be used to track crack growth over time, making it a practical tool for real-world infrastructure monitoring.

Keywords: Multi-scale deep learning · hairline crack detection · CNN–Transformer hybrid · crack evolution monitoring · structural health monitoring

1 Introduction

The long-term health of concrete structures often depends on catching problems early, and hairline cracks are one of the first signs of trouble. Identifying these tiny cracks is essential, but manual inspections are slow, subjective, and often inconsistent, especially in difficult field conditions.

Deep learning, particularly with models like U-Net [1] and DeepCrack [2], has offered a path to automated, pixel-perfect crack detection. These CNN-based methods are excellent at recognizing local patterns and textures. Their main drawback, however, is a limited field of view; they struggle to see the "big picture," making it hard for them to trace long, faint, or discontinuous cracks.

More recently, Transformer models like the Swin Transformer [3] have shown an ability to capture this missing global context using self-attention. But this

strength comes at a cost: they tend to smooth over the very fine details needed for precise boundary detection and often require huge datasets to work well.

This leaves us with a trade-off between local precision and global understanding. Our work, **CrackFormer**, aims to resolve this by combining the strengths of both approaches in a single hybrid model.

References

1. O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” *MICCAI*, pp. 234–241, 2015.
2. Q. e. a. Zou, “Deepcrack: Learning hierarchical feature representations for crack detection,” in *IEEE IPAS*, 2018, pp. 1–6.
3. Z. e. a. Liu, “Swin transformer: Hierarchical vision transformer using shifted windows,” in *ICCV*, 2021, pp. 10 012–10 022.